Marmara Lectures

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# Periodic dirac deltas

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## Periodic Dirac Delta Functions

Consider the functions

$$P_1(t) = \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}$$
$$P_2(t) = \frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})}$$

$P_1(t)$ is periodic with a period $2\pi$.

$P_2(t)$ is an odd function, $P_2(t) = P_2(-t)$, independently of $N$. Therefore

$$\int_{-A}^{A} \frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} dt = 0$$

for any $A \in \mathbb{R}$.

The integral of $P_1(t)$ for a single period is independent of $N$

$$\begin{aligned}\frac{d}{dN}\int_{-\pi}^{\pi} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} dt &= (N+\frac{1}{2})\int_{-2\pi}^{2\pi} \frac{\cos[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} dt \\ &= (N+\frac{1}{2})\int_{-2\pi}^{2\pi} P_2(t)dt = 0\end{aligned}$$

which follows from the oddness of $P_2(t)$.

**Theorem:**

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi \sum_{k=-\infty}^{\infty} \delta(t-2\pi k)$$

**Proof:** As P_1 (t) is periodic with period $2\pi$, it is sufficient to prove the theorem for a single period. For the interval $-\pi < t < \pi$ the above theorem reduces to

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi\delta(t)$$

Let $f(t)$ be a smooth and bounded function for $-\pi < t < \pi$. Foe $-\pi < a < 0 < b < \pi$ consider the integral

$$\int_a^b \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt$$

+

we can separate this integral into three parts:

$$\int_a^{-\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt + \int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt + \int_{\epsilon}^{b} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt$$

- The first and third integrals are multiplications of a smooth function $\frac{f(t)}{\cos(\frac{t}{2})}$ with an infimitely fast oscillation $\lim_{N\to\infty} \sin[(N+\frac{1}{2})t]$. By the Riemann-Lebesgue theorem, the result is zero.
- The second integral is taken in a very narrow region around zero. In this region
  $-f(t) \approx f(0)$
  $-\sin(\frac{t}{2}) \approx \frac{t}{2}$

therefore

$$\int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = f(0) \int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\frac{t}{2}} dt$$

but we have derived above that

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\frac{t}{2}} = 2\pi\delta(t)$$

therefore

$$\int_{-\epsilon}^{\epsilon} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = f(0)\int_{-\epsilon}^{\epsilon} 2\pi\delta(t)dt = 2\pi f(0)$$

and

$$\int_{a}^{b} \lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} f(t)dt = 2\pi f(0)$$

This indicates that

$$\lim_{N\to\infty} \frac{\sin[(N+\frac{1}{2})t]}{\sin(\frac{t}{2})} = 2\pi\delta(t)$$

on a single period. When we extend this result to all the periods, we prove the theorem.

**Theorem:**

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